

Apache Web Server Log Analysis Report

Security Monitoring and Threat Detection Using Apache Access and Error Logs

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Project Type: Blue Team / SOC Lab

Environment: Ubuntu Linux, Apache2, Access Logs, Error Logs

Duration: June 2026

Executive Summary

This project demonstrates practical SOC skills through Apache web server log analysis. Access and error logs were analyzed to identify web traffic patterns, HTTP 404 responses, suspicious URL requests, and server operational events. The project simulates real-world SOC analyst activities including monitoring, investigation, threat detection, reporting, and MITRE ATT&CK; mapping.

Project Objectives

- Install and configure Apache Web Server
- Generate web traffic
- Analyze access and error logs
- Identify failed HTTP requests
- Investigate web traffic patterns
- Produce a professional incident investigation report

Lab Environment

Operating System: Ubuntu Linux

Web Server: Apache2

Log Location: /var/log/apache2

Analysis Tools: grep, awk, less, sort, uniq

Key Findings

Finding 1: Multiple HTTP 404 responses observed for /admin, /login, /test, /secret and /favicon.ico.

Finding 2: Top source IP was 127.0.0.1 with 22 requests.

Finding 3: Homepage '/' was the most requested resource.

Finding 4: No critical Apache service errors were detected.

MITRE ATT&CK; Mapping

T1595 Active Scanning

T1590 Gather Victim Network Information

T1083 File and Directory Discovery

T1082 System Information Discovery

T1580 Infrastructure Enumeration

SOC Analyst Investigation Notes

Indicators observed: Suspicious URLs detected, HTTP 404 responses detected, privileged endpoint access attempts detected. No malware activity, service failure, or data exfiltration observed.

Security Recommendations

Monitor recurring 404 errors, implement SIEM alerting, review web logs regularly, restrict administrative interfaces, enable centralized logging, and deploy a web application firewall.

Skills Demonstrated

Log Analysis, Security Monitoring, Threat Detection, Event Investigation, Security Reporting, Linux Administration, Apache Administration, MITRE ATT&CK; Mapping, Incident Investigation.

Conclusion

The project successfully demonstrated foundational SOC analyst skills including log analysis, threat detection, web server monitoring, and security reporting. The investigation mapped observed behaviors to MITRE ATT&CK; techniques and produced actionable security recommendations.